

Blind Image Quality Assessment By Limited Progressive Learning

Qingyuan Wang

Peking University

2200013187@stu.pku.edu.cn

Abstract—Blind Image Quality Assessment (BIQA) is essential for evaluating image quality without the need for reference images. In this paper, we propose a novel approach inspired by the human learning process, where the model progressively learns increasingly complex tasks from simpler ones. Our main contribution lies in designing a new network architecture for BIQA that incrementally leverages knowledge from simpler tasks to tackle progressively harder challenges. Specifically, as the tasks become more complex, the model incorporates increasingly detailed features from shallower layers, which are crucial for fine-grained quality assessment. Experimental results demonstrate that our model outperforms existing state-of-the-art methods, showcasing its superior effectiveness in BIQA.

Index Terms—BIQA, Progressive Learning, Limited knowledge

I. INTRODUCTION

Image Quality Assessment (IQA) is a fundamental component in various image processing and computer vision applications, such as object recognition [3] and scene classification [4]. Accurate IQA is essential for ensuring that images meet desired quality standards without introducing perceptible artifacts. Traditional IQA methods often rely on the availability of high-quality reference images to compare against distorted ones (FRIQA) [2]. However, in many real-world scenarios, reference images are unavailable, making Blind Image Quality Assessment (BIQA) [1] increasingly important. BIQA aims to predict the perceptual quality of images without any reference, aligning more closely with practical applications where only the distorted images are accessible. Despite its significance, the development of effective BIQA methods is constrained by the limited size of available datasets, necessitating efficient and robust approaches that can generalize well across different types of distortions.

Current BIQA methods have made progress but still face significant challenges. Many existing models struggle to achieve satisfactory performance, particularly when applied to real-world datasets with authentic distortions. The lack of generalization to unseen distortions and diverse image content limits their applicability in practical settings. Moreover, the performance gap between BIQA models and human perception remains substantial, indicating that current methods do not fully capture the complex factors influencing perceived image quality.

Inspired by the human learning process, where individuals build upon foundational knowledge to tackle more complex tasks, we propose a progressive learning approach for BIQA.

Humans often learn by mastering simple concepts before moving on to more challenging ones, effectively transferring knowledge from basic tasks to facilitate the understanding of intricate details. By simulating this learning strategy, we aim to enhance the model's ability to assess image quality more accurately. Specifically, we focus on real distortion datasets to ensure that our method is applicable to practical scenarios involving authentic image degradations.

In this paper, we introduce a novel network architecture for BIQA that leverages a series of progressively more difficult sub-tasks designed through quality classification. Our approach incrementally incorporates detailed features extracted from shallower layers of the network as the tasks become more complex. In simpler tasks, we limit the use of these detailed features to encourage the model to focus on more general quality assessments. This progressive learning strategy allows the model to build upon the knowledge gained from simpler tasks to improve performance on more challenging assessments. Experimental results demonstrate that our method outperforms several state-of-the-art BIQA models, showcasing superior effectiveness and generalization capabilities.

II. OUR APPROACH

A. Subtask design

In the process of progressive learning, it is essential to define tasks with varying levels of difficulty. While some approaches attempt to design new tasks to assist IQA task [10], we opt to use the quality estimation task itself as the subtasks, drawing inspiration from PMT-IQA [7]. Specifically, we partition the quality range into n subintervals of equal length, transforming our subtasks into quality classification tasks.

$$\mathcal{D}_i \in \text{Category } k \quad \text{if} \quad \left\lfloor \frac{i - \mathcal{Y}_{\min}}{(\mathcal{Y}_{\max} - \mathcal{Y}_{\min})/n} \right\rfloor = k \quad (1)$$

Here, $\mathcal{D}$ represents the dataset, and $\mathcal{D}_i$ is the i -th image. $\mathcal{Y}_{\max}$ and $\mathcal{Y}_{\min}$ denote the maximum and minimum quality scores of the dataset $\mathcal{D}$, respectively.

A larger n results in a more challenging subtask. This is intuitive because a smaller n only requires a rough estimation of image quality. For instance, when n equals 2, the task reduces to determining whether the image quality is "good" or "bad." As n increases, the network is required to predict quality scores with greater precision.

This subtask design allows the network to progressively acquire skills, from the basic ability to classify an image

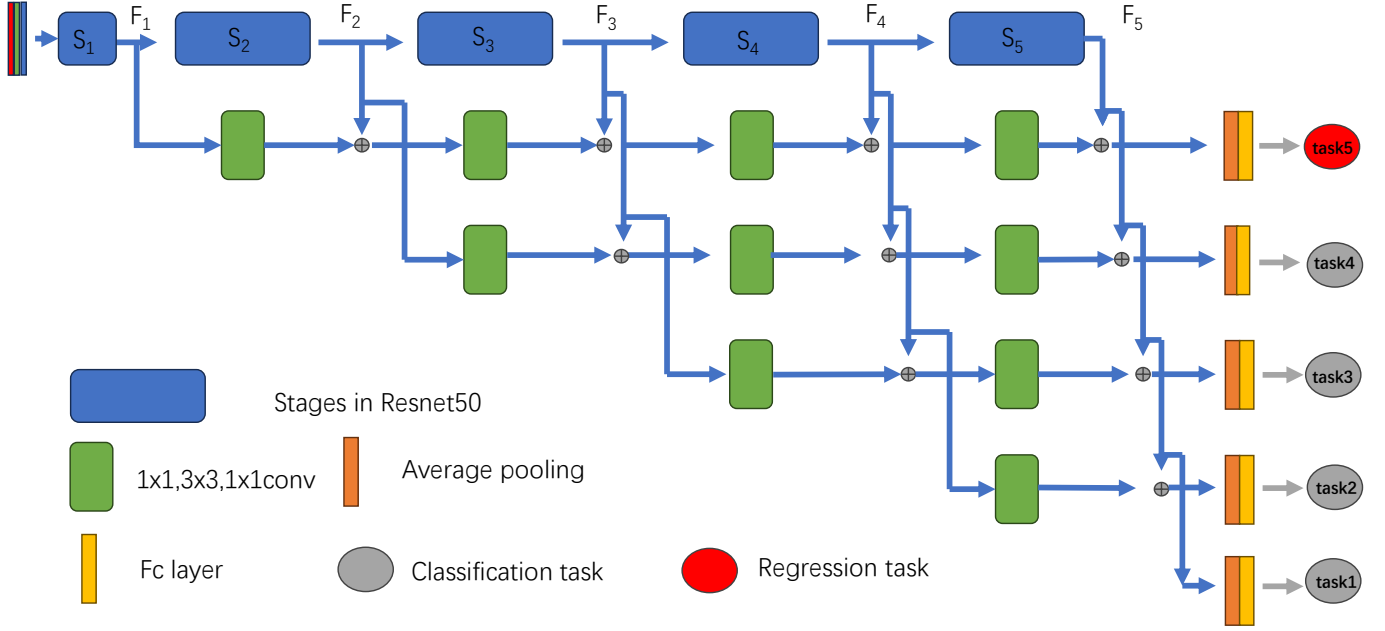


Fig. 1. Network Architecture

as ”good” or ”bad” to the advanced capability of predicting precise quality scores.

In this paper, we use four classification subtasks, with the number of categories set to 2, 4, 6, and 8, respectively. Experimental results show that if n is too large, the task exceeds the model’s capability, leading to degraded performance. Conversely, if n is too small, the knowledge learned is overly simplistic, resulting in suboptimal outcomes.

B. Network Architecture

Our network architecture is illustrated in Figure 1. The core idea of our model is to leverage feature maps extracted from different stages of the backbone model to solve progressively more difficult tasks. The proposed model consists of two main components: the feature extraction backbone and the feature fusion network.

In this paper, we use ResNet50 [8] as the feature extraction backbone. ResNet50 is divided into five stages, denoted as $S_1, S_2, \dots, S_5$, and the output of each stage is taken as the feature map for that stage, denoted as $F_1, F_2, \dots, F_5$. These feature maps are used to handle tasks of increasing difficulty.

For feature fusion, inspired by StairIQA [9], each task has its own ”path”, which combines feature maps from the relevant stages. Since feature maps from different stages may have different sizes, we use a transformation function W to adjust the feature map dimensions to be compatible for fusion.

Specifically, W is a block of three convolutional layers (1x1, 3x3, 1x1) that scales each feature map to a consistent size:

$$WF_i = \text{Conv3}(\text{Conv2}(\text{Conv1}(F_i))), \quad (2)$$

where Conv1, Conv2, and Conv3 represent the 1x1, 3x3, and 1x1 convolutional layers, respectively. This structure ensures that feature maps from different stages can be fused together seamlessly.

As the tasks become more complex, the length of each path increases by incorporating feature maps from preceding stages. For task t , the feature map is constructed as:

$$F_t = (WF_{t-1}) \oplus F_{t-2}, \quad (3)$$

where $\oplus$ represents the element-wise plus.

After the feature fusion and convolution operations, we apply average pooling to each path’s output, followed by a fully connected (FC) layer. Finally, a softmax activation function is applied to the classification tasks, and a regression head is used for the regression task.

As shown in Figure 1, task 5 is a regression task, while tasks 1 through 4 are classification tasks.

For simpler tasks, we prefer to use deeper features instead of shallow features, as deeper features are more high-level and contain more abstracted information. In quality prediction tasks, deeper features play a crucial role because they encapsulate global patterns and semantic information, which are essential for evaluating overall image quality. In contrast, shallow features, which capture more fine-grained details like

Database	BID		LIVEC		Koniq		SPAQ		FLIVE		FLIVE_patch	
	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
QAC	0.0686	0.0138	0.3258	0.3229	0.3430	0.2961	0.0465	0.1072	0.1042	0.0656	0.1720	0.1042
NIQE	0.4536	0.4676	0.4772	0.4713	0.5260	0.4745	0.6973	0.6850	0.1048	0.1409	0.3211	0.2826
ILNIQE	0.4531	0.5114	0.4946	0.4538	0.5029	0.4956	0.7194	0.6540	0.2188	0.2547	0.5306	0.5231
BRISQUE	0.6005	0.6211	0.5736	0.5401	0.7150	0.7016	0.8021	0.8056	0.3201	0.3561	0.5372	0.5843
BMPRI	0.4868	0.5229	0.5154	0.4583	0.6577	0.6546	0.7501	0.7544	0.2737	0.3146	0.5839	0.6142
CNNIQA	0.6269	0.6008	0.6163	0.6144	0.6852	0.6837	0.7959	0.7988	0.3059	0.2850	0.6005	0.5379
WaDIQaM-NR	0.6916	0.7304	0.6526	0.6359	0.7294	0.7538	0.8397	0.8449	0.4346	0.4303	0.6995	0.7197
SFA	0.8037	0.8213	0.8202	0.8253	0.8882	0.8966	0.9057	0.9069	0.5415	0.6260	0.7175	0.7501
DB-CNN	0.8443	0.8624	0.8450	0.8590	0.8780	0.8867	0.9099	0.9133	0.5537	0.6518	0.7509	0.7869
HyperIQA	0.8546	0.8709	0.8544	0.8585	0.9075	0.9205	0.9155	0.9188	0.5354	0.6228	0.7489	0.7850
UNIQUE	0.8540	0.8900	0.8580	0.8730	0.8960	0.9010	-	-	-	-	-	-
StairIQA	0.8799	0.8901	0.9028	0.9242	0.9148	0.9317	0.9210	0.9253	0.5634	0.6839	0.7769	0.8012
proposed	0.8843	0.9011	0.9077	0.9259	0.9038	0.9206	0.9212	0.9272	0.5634	0.6638	0.7801	0.8106

TABLE I

Test dataset	methods	BID	LIVEC	Koniq10k	SPAQ	FLIVE	FLIVE_patch
BID	StairIQA	\	0.8882	0.8767	0.8434	0.8488	0.7969
	Proposed	\	0.8970	0.8776	0.8773	0.8534	0.8403
LIVEC	StairIQA	0.8603	\	0.8450	0.8369	0.8510	0.8007
	Proposed	0.8634	\	0.8512	0.8707	0.8490	0.8439

TABLE II

	BID SRCC	BID PLCC	LIVEC SRCC	LIVEC PLCC
No	0.8801	0.8945	0.8844	0.9032
Reverse	0.8745	0.8857	0.8969	0.9188
Fixed weight	0.8791	0.8932	0.9023	0.9216
Proposed	0.8843	0.9011	0.9077	0.9259

TABLE III

textures, can enhance precision in more challenging tasks but are less critical for general quality assessment.

In addition, we employed a multi-dataset training strategy and made improvements based on StairIQA. While StairIQA uses six separate regressors and trains each dataset one epoch at a time, we adopted a mix-batch approach, where data from multiple datasets are included within a single batch. This ensures that the training process is more fair and balanced.

C. Loss Functions

We use the mean squared error(MSE) loss for regression tasks.

$$\mathcal{L}_{\text{MSE}}(x|\theta) = \frac{1}{n} \|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 \quad (4)$$

Where $\mathbf{y}$ is the ground truth score, $\hat{\mathbf{y}}$ is the predicted score. Then we use the cross entropy(CE) loss for classification tasks.

$$\mathcal{L}_{\text{CE}}(x|\theta) = -\frac{1}{n} \sum_{i=1}^n \mathbf{y}_i^T \log(\hat{\mathbf{y}}_i) \quad (5)$$

$\mathbf{y}_i$ denotes the truth label for category i on data x , as we defined before.

Instead of setting a fixed weight for subtasks, we follow dynamic weight method from [10]. The weight vector can be

calculated automatically according to the relative descending rate.

III. EXPERIMENT

In this section, we describe the experiment setting first and then report the experiment result between our proposed model and state-of-art model. Ablation studies are conducted to validate the effectiveness of our method.

A. Experimental Setups

Our experiment is mainly conducted on six authentically distorted datasets: BID [11], CLIVE [12], SPAQ [13], FLIVE [14] (including both FLIVE and FLIVE Patch), Koniq10k [15]. Specifically, We randomly sample 80% images from each dataset as the training set, leaving the other 20% images as test set. We repeat this procedure for 3 times, and we report the mean of Spearman rank-order correlation coefficient (SRCC) and Pearson linear correlation coefficient (PLCC). SRCC and PLCC is computed as follows.

$$\text{SRCC} = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i represents the difference between the ranks of the two variables for the i -th sample, and n is the number of samples.

$$\text{PLCC} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

where x_i and y_i are the values of the two variables, and $\bar{x}$ and $\bar{y}$ are the means of x_i and y_i , respectively.

B. Main Comparison

We compared our proposed methods with various kinds of state-of-art BIQA models, including: 1) Handcrafted feature based BIQA models: QAC [16], NIQE [17], ILNIQE [18], BRISQUE [19], and BMPRI [20]. 2) Deep learning based models: CNNIQA [21], WaDIQaM-NR [22], SFA [23], DB-CNN [24], HyperIQA [25], UNIQUE [26], and StairIQA [9]. The results are shown in table1.

Results show that Our model out-performed other models on datasets except Koniq.

C. Cross-dataset Evaluation

In this section, we compare the generalizability of our proposed model by training and testing on different dataset. We compared our proposed model to StairIQA [9], which out-performs other methods. The results are in table2. We can see that our proposed method shows a excellent generalizability between different datasets.

D. Ablation Study

To test if the method we proposed has influence on the performance of the model, we conduct ablation study. We test the result on BID and LIVEC. The results are in table 3. "No" represents we remove the subtasks and only keep the regression task. "Reverse" represents we reverse the order of classification task(task1 to task4). "Fixed weight" means we fix the weight of loss from classification tasks and regression task to the same value 1.

The results show that our training methods is effective. By comparing the "No" and "Fixed weight" methods, we can see that even without dynamic weighting, the performance on the BID dataset is similar. However, on the LIVEC dataset, our learning approach achieves an improvement of nearly 2%.

Here, we can observe that on the BID dataset, the "No" method performs better than the "Reverse" method, while the opposite is true for the LIVEC dataset. We explain this by noting that although the "Reverse" method introduces additional tasks for training, it does not follow the learning direction from easy to difficult. As a result, it slightly undermines the performance. This also validates the effectiveness of our approach to limited progressive learning.

IV. CONCLUSION

In conclusion, this paper introduces a novel Blind Image Quality Assessment (BIQA) approach that emulates the human learning process through progressive task learning. We specifically designed a series of score classification tasks with varying levels of difficulty, where the model incorporates an increasing amount of low-level features as task complexity grows. This design allows for more accurate score predictions.

Experimental results validate the effectiveness of our method, demonstrating its superior accuracy and generalization compared to state-of-the-art techniques. Our approach not only enhances BIQA performance but also emphasizes the importance of progressively leveraging detailed features for fine-grained quality assessments.

ACKNOWLEDGMENT

I would like to express my deepest gratitude to Professor Tingting Jiang, whose invaluable guidance and insights were crucial to the completion of this research. I am also thankful to my colleagues Yan Zhong for his guidance and constructive feedback throughout the research process.

REFERENCES

- [1] Zhou Wang and Alan C. Bovik. Modern Image Quality Assessment. Morgan & Claypool, 2006.
- [2] Zhou Wang, Alan C. Bovik, Hamid R Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, Apr. 2004.
- [3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009.
- [4] Matthew R. Boutell, Jiebo Luo, Xipeng Shen, and Christopher M. Brown. Learning multi-label scene classification. *Pattern Recognition*, 37(9):1757–1771, Sep. 2004.
- [5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].
- [7] Q Pan, N Guo, L Qingge, J Zhang, P Yang. PMT-IQA: progressive multi-task learning for blind image quality assessment. 20th Pacific Rim International Conference on Artificial Intelligence, 2023.
- [8] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, Jun.2016.
- [9] W Sun, X Min, D Tu, S Ma, G Zhai. Blind quality assessment for in-the-wild images via hierarchical feature fusion and iterative mixed database training. *IEEE Journal of Selected Topics in Signal Processing*. 2023.
- [10] Weixia Zhang, Guangtao Zhai, Ying Wei, Xiaokang Yang, and Kede Ma. Blind image quality assessment via vision-language correspondence: A multitask learning perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14071–14081, 2023.
- [11] A. Ciancio, A. L. N. T. da Costa, E. A. B. da Silva, A. Said, R. Samadani, and P. Obrador, "No-reference blur assessment of digital pictures based on multifeature classifiers," *IEEE Transactions on Image Processing*, vol. 20, no. 1, pp. 64–75, 2011.
- [12] D. Ghadiyaram and A. C. Bovik, "Massive online crowdsourced study of subjective and objective picture quality," *IEEE Transactions on Image Processing*, vol. 25, no. 1, pp. 372–387, 2016.
- [13] Y. Fang, H. Zhu, Y. Zeng, K. Ma, and Z. Wang, "Perceptual quality assessment of smartphone photography," in 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 3677–3686.
- [14] Z. Ying, H. Niu, P. Gupta, D. Mahajan, D. Ghadiyaram, and A. Bovik, "From patches to pictures (paq-2-piq): Mapping the perceptual space of picture quality," in 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 3575–3585.
- [15] Z. Ying, H. Niu, P. Gupta, D. Mahajan, D. Ghadiyaram, and A. Bovik, "From patches to pictures (paq-2-piq): Mapping the perceptual space of picture quality," in 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 3575–3585.
- [16] W. Xue, L. Zhang, and X. Mou, "Learning without human scores for blind image quality assessment," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 995–1002.

- [17] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a "completely blind" image quality analyzer," *IEEE Signal processing letters*, vol. 20, no. 3, pp. 209–212, 2012.
- [18] L. Zhang, L. Zhang, and A. C. Bovik, "A feature-enriched completely blind image quality evaluator," *IEEE Transactions on Image Processing*, vol. 24, no. 8, pp. 2579–2591, 2015.
- [19] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Transactions on image processing*, vol. 21, no. 12, pp. 4695–4708, 2012.
- [20] X. Min, G. Zhai, K. Gu, Y. Liu, and X. Yang, "Blind image quality estimation via distortion aggravation," *IEEE Transactions on Broadcasting*, vol. 64, no. 2, pp. 508–517, 2018.
- [21] L. Kang, P. Ye, Y. Li, and D. Doermann, "Convolutional neural networks for no-reference image quality assessment," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1733–1740.
- [22] S. Bosse, D. Maniry, K.-R. Muller, T. Wiegand, and W. Samek, "Deep neural networks for no-reference and full-reference image quality assessment," *IEEE Transactions on Image Processing*, vol. 27, no. 1, pp. 206–219, 2017.
- [23] D. Li, T. Jiang, W. Lin, and M. Jiang, "Which has better visual quality: The clear blue sky or a blurry animal?" *IEEE Transactions on Multimedia*, vol. 21, no. 5, pp. 1221–1234, 2018.
- [24] W. Zhang, K. Ma, J. Yan, D. Deng, and Z. Wang, "Blind image quality assessment using a deep bilinear convolutional neural network," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 30, no. 1, pp. 36–47, 2020.
- [25] S. Su, Q. Yan, Y. Zhu, C. Zhang, X. Ge, J. Sun, and Y. Zhang, "Blindly assess image quality in the wild guided by a self- adaptive hyper network," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3667–3676.
- [26] W. Zhang, K. Ma, G. Zhai, and X. Yang, "Uncertainty-aware blind image quality assessment in the laboratory and wild," *IEEE Transactions on Image Processing*, vol. 30, pp. 3474–3486, 2021.